Power

M2
Module-power_12and5V-0.2

+12V_FROM_MAIN_RELAY
IN_CONSTANT_POWER
IN_VIGN
PWR_EN
+12V
VBAT
VIGN
+5V
+5VP
PG_5VP
GND

Bus types
General purpose bus
Ignition
Injection
Analog inputs (sensors)
Discrete inputs
LS & HS outputs
DC Motor

M4
Module-motor-driver-0.1
DC driver
M7
Module-motor-driver-0.1

OUT_ETB1
OUT_ETB2
OUT_WASTEGATE1
OUT_WASTEGATE2
ETB_DIS
ETB_PWM
ETB_DIR
WASTEGATE_DIS
WASTEGATE_PWM
WASTEGATE_DIR
SPI3_CS_ETB
SPI3_CS_WASTEGATE

M5
Module-mcz33810-0.1

OUT_IGN1
OUT_IGN2
OUT_IGN3
OUT_IGN4
OUT_INJ1
OUT_INJ2
OUT_INJ3
OUT_INJ4
IGN1
IGN2
IGN3
IGN4
INJ1
INJ2
INJ3
INJ4
SPI3_MOSI
SPI3_SCK
SPI3_CS_33810
SPI3_MISO
-OUTEN
-SPIKOUR
NOMI
MAXI

LS4 FLYBACK1
Sidehold capable
OUT_CVVT1
OUT_CVVT2
OUT_WGT
OUT_RCV
OUT_AC_RELAY
OUT_AUX_LS1
OUT_AUX_LS2
OUT_CPSV_EVAP_PURGE
OUT_HEATER_CONTROL
12V_FLYBACK
CVVT1
CVVT2
WGT
RCV
AC_RELAY
AUX_LS1
AUX_LS2
CPSV_EVAP_PURGE
HEATER_CONTROL

LS6
Always hot load capable
OUT_CHECK_ENGINE
OUT_TACH
OUT_MAIN_RELAY
OUT_FAN_RELAY_LOW
OUT_FAN_RELAY_HIGH
OUT_FUEL_PUMP_RELAY
CHECK_ENGINE
TACH
MAIN_RELAY
FAN_RELAY_LOW
FAN_RELAY_HIGH
FUEL_PUMP_RELAY

M1
Module-mega-mcu144-0.6

SPI2_SCK/CAN2_TX
SPI2_MISO
SPI2_MOSI
SPI2_CS/CAN2_RX
SPI3_SCK
SPI3_MISO
SPI3_MOSI
SPI3_CS
CANH
CANL
I2C_SDA
I2C_SCL
UART8_RX
UART8_TX
UART2_RX
UART2_TX
USBP
USBM
USBID
VBUS
SWCLK
SWDIO
SWO
nReset
BOOT0
SPI3_CS_33810
CPSV_EVAP_PURGE
HEATER_CONTROL
CHECK_ENGINE
FAN_RELAY_LOW
FAN_RELAY_HIGH
FUEL_PUMP_RELAY
AC_RELAY
MAIN_RELAY
WASTEGATE_DIR
WASTEGATE_PWM
ETB_DIR
ETB_PWM
WASTEGATE_POS
IN_CRANK
IN_CAM
IN_VSS
IN_TPS
IN_TPS2
IN_PPS
IN_PPS2
IN_IAT
IN_ECT
IN_AT2
IN_OIL
IN_AC_ON
IN_CLUTCH
IN_BRAKE
IN_MAP1
IN_MAP2
IN_MAP3
IN_AC_PRESSURE
IN_WBO
IN_BK1_O2_NARROW
IN_WASTEGATE_POS
IN_PS_SWITCH
DIN1
DIN2
DIN3
KNOCK
IN_KNOCK
IN_CAM1
IN_CAM2
IN_VSS
PWR_EN
VIGN
VBAT
OUT_PWR_EN
VREF1
VREF2
V33_REF
IN_D1
IN_D2
IN_D3
IN_D4
OUT_PWM1
OUT_PWM2
OUT_PWM3
OUT_PWM4
OUT_PWM5
OUT_PWM6
OUT_PWM7
OUT_PWM8
OUT_IO1
OUT_IO2
OUT_IO3
OUT_IO4
OUT_IO5
OUT_IO6
OUT_IO7
OUT_IO8
OUT_IO9
OUT_IO10
OUT_IO11
OUT_IO12
OUT_IO13
+3.3VA
+3V3
+5VA
-SPIKOUR
MAXI
NOMI
-OUTEN
IN_CRANK
IN_CAM2
IN_CAM1
IN_CLUTCH
IN_BRAKE
IN_MAP1
IN_MAP2
IN_AC_PRESSURE
IN_WBO
IN_BK1_O2_NARROW
IN_WASTEGATE_POS
IN_PS_SWITCH
DIN1
DIN2
DIN3
IN_KNOCK
IN_CAM1
IN_CAM2
IN_VSS
VREF1
VREF2
V33_REF
OUT_PWR_EN
IN_VIGN
VBAT
VCC
GND

M6
Module-can-0.1
M5
Module-wbo-0.5

CAN2_RX
CAN2_TX
CAN_VIO
CAN_RX
CAN_TX
CANL
CANH
+5V
+3V3
GND
CAN-
CAN+
Power GND
Heater negative
V33_OUT
SWCLK
SWDIO
nReset
VDDA
VS_IN
LSU_H-
LSU_H+
LSU_Rtrim
LSU_Ip
LSU_Vm
LSU_Un
PULL_UP1
SEL1
PULL_DOWN1
PULL_UP2
SEL2
PULL_DOWN2
+5VA
GND
WBO_Heater
WBO_Rtrim
WBO_Ip
WBO_Vm
WBO_Un
+12V_FROM_MAIN_RELAY
39281063

USB_B
J40
Solid
Conn_01x05_Male

P1B
284617-1

P1A
284617-1

IN_VIGN
IN_CONSTANT_POWER
IN_MAP2
IN_KNOCK_RAW
IN_CAM2
IN_CRANK
IN_TEMP
IN_WASTEGATE_POS
OUT_IGN1
OUT_IGN2
OUT_IGN3
OUT_IGN4
IN_MAP1
IN_TPS2
IN_CLT
IN_PPS2
IN_VSS
IN_BRAKE
OUT_IGN1
IN_CLUTCH
OUT_WASTEGATE2
OUT_IGN3
IN_AC_ON
IN_PS_SWITCH
OUT_WASTEGATE1
OUT_TACH
OUT_AC_RELAY
OUT_FAN_RELAY_HIGH
OUT_MAIN_RELAY
OUT_FAN_RELAY_LOW
OUT_CVVT1
OUT_CPSV_EVAP_PURGE
OUT_CVVT2
OUT_FUEL_PUMP_RELAY
OUT_ETB2
OUT_ETB1
IN_CAM1
IN_WBO
IN_TPS1
IN_PPS1
IN_OIL
IN_IAT
IN_AC_PRESSURE
OUT_HEATER_CONTROL
CAN+
CAN-
GND
+5VP
+12V_FROM_MAIN_RELAY
P1 O2
P9 Alternator
P10 VIS
P11 Blower motor
P13 IMMO
P14
P15 IMMO
P16 DATA
P17 CRUISE
P19 CRUISE
P20
P22 O2
OEM_BK1
used by Kia forte
Kia Sg Kia Forte
P23
P24
P25
P26
P27
P28
P29
P30 Wiper SW
P31 Generator FR
P32
P33
P34
P35
P36
P37
P38
P39 cruise
P40
P41
P42
P43
P44
P45
P46
P47
P48
P49 Brake test
P50
P51
P52
P53
P54
P55
first generation BK1

J2
39281123

OUT_AUX_LS1
OUT_AUX_LS2
DIN1
DIN2
DIN3
AIN1
AIN2
AIN3

R1 R2 R3 R4 R5 R6 R7 R8 R9 R10 R11 R12 R13 R14 R15 R16 R17 R18 R19 R20 R21 R22 R23 R24 DNP

LEDs

R6 10k
D6 D7 D8 D9 LED-RED-0603
IN4 IN3 IN2 IN1
R7 20k
R76 20k
D16 D17 D19 D18 D36 D37 D38 D35 LED-WHITE-0603
1N4148W
OUT_TACH
OUT_FAN_RELAY_LOW
OUT_MAIN_RELAY
OUT_AC_RELAY
OUT_FUEL_PUMP_RELAY
OUT_CHECK_ENGINE
OUT_FAN_RELAY_HIGH
+5V
R10 20k
D28 D29 D30 D27
IN_CRANK
IN_CAM2
IN_CAM1

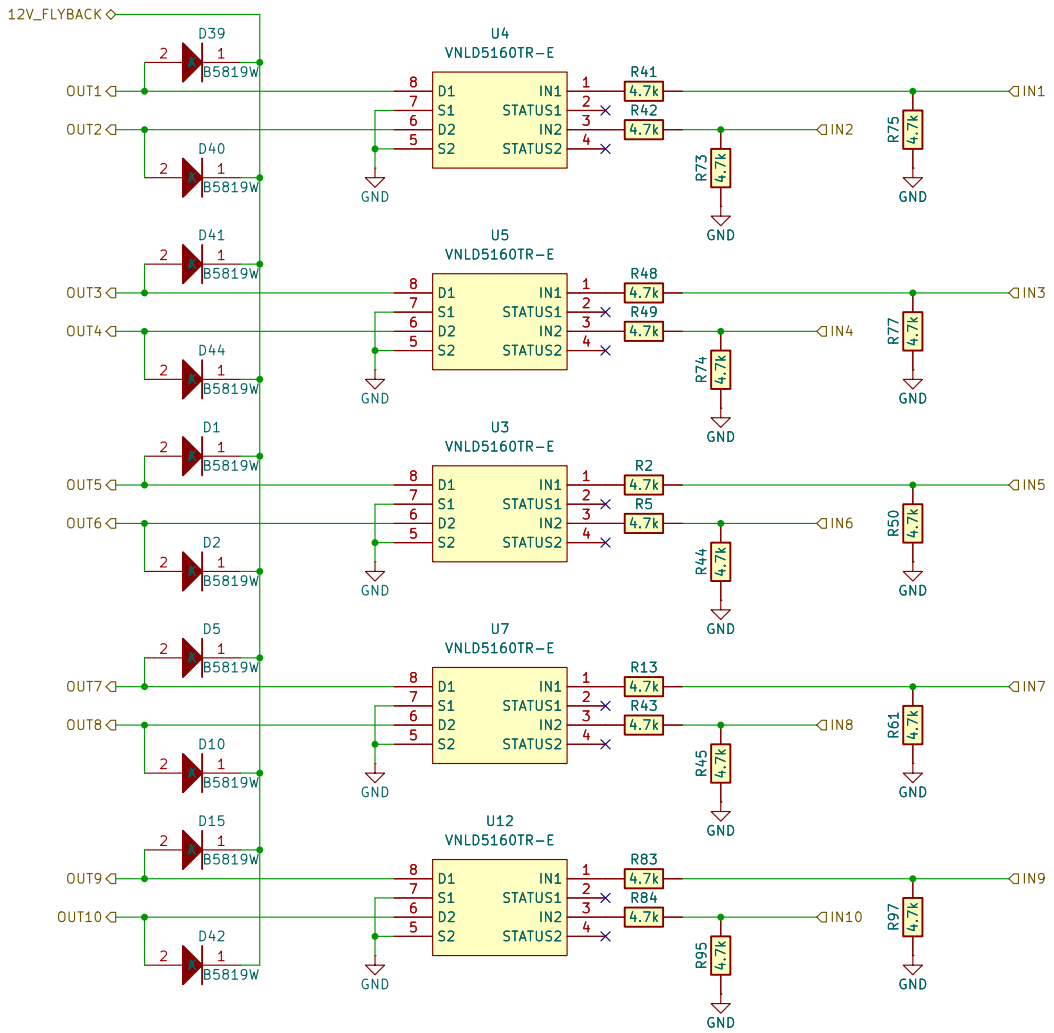
Prototype areas

J3
Conn_01x04
TX1
RX1
TX2
RX2
PROTO_AREA_2
J22
J24

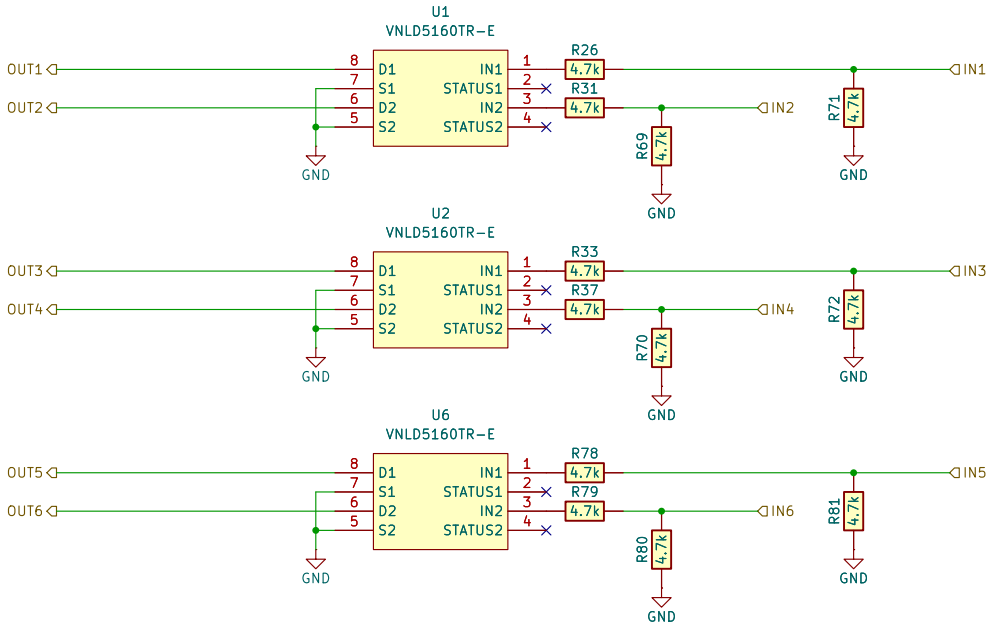
FOR OFF ROAD PURPOSES ONLY
This is not for applications with emissions or safety regulations (AKA not for street use). This is for closed stages, track and equipment.
https://rusefi.com/s/hellen154hyundai

Board Info
Hellen-One-PCB-logo

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Size: A1 | Date: 2024-05-13 | Rev: E
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